AUTO: before the match

Team 37102 · BioBuzz AUTO RED / BLUE (Pedro) · 30 s, hands off

1 🤝 Talk to partner
Who has S1, who parks in A / B, the GARDEN, our lane (yellow box).

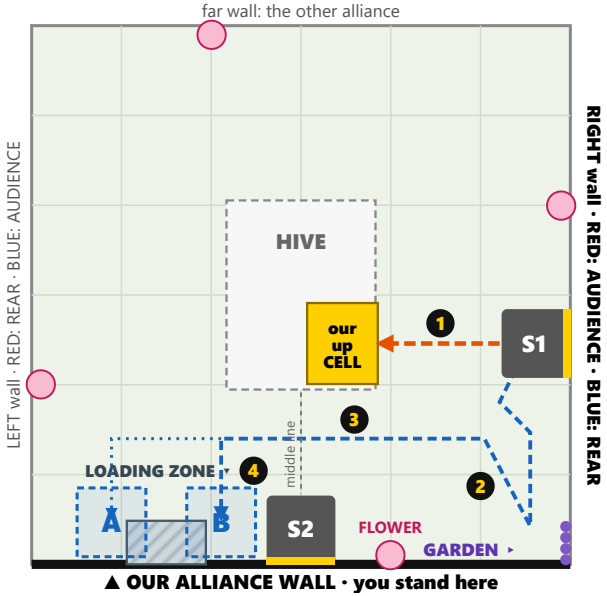
2 📋 Pick op mode
BioBuzz AUTO RED / **BLUE** = our colour (no other AUTO). TeleOp is preselected.

3 📍 Place + load
Robot at **S1** or **S2** (map). **4 pre-loads**, one in each windmill pocket.

4 ✋ INIT, hands off
Don't touch the robot until START: the Pinpoint calibrates while it is still.

5 🎮 Check + START
Set gamepad 1 buttons, read the screen. ▶ **START** when the match starts.

📍 Where to place it



▲ **OUR ALLIANCE WALL · you stand here**

Seen from where you stand: the same for RED and BLUE. Yellow bar = intake (front); launcher + camera at the back. **1** shoot · **2** GARDEN · **3** lane beside the HIVE · **4** park in B (dotted: to A)

S1 · CELL wall (default, best): **intake against the wall on your RIGHT** (RED: audience wall, BLUE: rear wall), **centred on our up CELL** (robot middle in line with the CELL middle), square to the wall. It shoots from here.

S2 · ALLIANCE wall: intake against our wall, centred on the middle line (between the FLOWER and the LOADING ZONE), square to the wall. It drives about 5 s to the shot spot, so the GARDEN is short or skipped.

🎮 INIT buttons (gamepad 1)

Press	Changes	Default
D-pad ← or →	Start S1 ⇄ S2	**S1**
□ **Square**	Park slot B ⇄ A	**B**
✕ **Cross**	Shoot pre-loads	**ON**
△ **Triangle**	GARDEN	**ON**
○ **Circle**	Stretch	**OFF**
D-pad ↑ / ↓	Start delay +1 / −1 s	**0 s** (0–10)
R1 / L1	Park by +1 / −1 s	**24 s** (18–27)

✕ △ ○ switch ON ⇄ OFF. One press = one change. **Start delay**: it waits that long after START. **Park by**: the latest time it stops and goes to park. **Stretch**: shoot again after the GARDEN (page 2). The screen calls L1 / R1 "LB / RB". **Every INIT starts from these defaults**: after a new INIT, set them again.

🤝 Agree with your partner (every match)

1. **Start:** only one robot can use S1.
2. **Park:** one robot in A, one in B (the screen says `we park in slot B, they take slot A`). They park **only at their end** of the zone, close to the wall.
3. **GARDEN:** we take our GARDEN POLLEN. They should not.
4. **Lane:** at the end of AUTO they stay out of our lane beside the HIVE (map, 3).

B is the default: the way to A goes past B.

📺 INIT screen: a good one

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BIOBUZZ AUTO RED - Pedro                    our colour?
Start (D-pad L/R) : CELL wall: side wall in front of our up   as agreed
CELL (shoots from here)
Park slot (Square/X) : B = MIDDLE end of the LOADING          as agreed
ZONE (toward the FLOWER)
>> Tell your partner: we park in slot B, they take slot A
Shoot pre-loads (Cross/A) : ON                          ON
GARDEN POLLEN (Triangle/Y) : ON                         ON
Stretch: shoot again (Circle/B) : OFF
Start delay (D-pad U/D) : 0 s
Park by (LB/RB) : 24 s at the latest
Plan : shot spot + 4 pre-loads done ~6 s
Plan : GARDEN done ~17 s (intake on, 2 in back and forth ...)
Plan : PARK B starts by 23 s, takes ~6 s
Route check : OK                                        must be OK
Pedro : ready                                           must be ready
Camera : no CELL in view (normal at the wall)           ok
Place: intake against the AUDIENCE wall, centred on           robot there?
our up CELL, square to the wall
```

RED, all defaults. BLUE: AUTO BLUE, REAR wall. Plan times are estimates.

Bad line	What happens · what you do
`Pedro : NOT READY …`	No Pinpoint or Pedro not tuned: AUTO only backs 8 in off the wall (LEAVE, 3 points). Keep still; tell the coach.
`Camera : NOT FOUND …`	Fires the pre-loads unaimed, no stretch. Tell the coach.
`GARDEN POLLEN : ON but needs shooting (G407)`	Shoot is OFF, so no GARDEN. ✕ turns Shoot ON.
`Route check : PROBLEM …`	Tell the coach.
`GARDEN ON but no "GARDEN done" Plan line`	Not enough time: it will skip the GARDEN.

AUTO: in the match

what the robot does · hands off · if something goes wrong · checklist

What the robot does by itself (30 s)

1 0–1 s
LEAVE
Leaves the wall, spins up the flywheel.
LEAVE +3 (counted at the end)

2 1–7 s
SHOOT
Camera aims at our up CELL, fires the 4 pre-loads.
HIVE TIP +20 if it tips

3 7–14 s
GARDEN
Sweeps slowly along the wall into the GARDEN corner, intake on.
POLLEN for TeleOp

4 14–18 s
CORNER
Intake on up to 4 s, going 2 in back and forth, so the last POLLEN rolls in.

5 ~18 s
PARK
In our LOADING ZONE. Slot B: parked by ~24 s. Slot A: by ~28 s.
PARK +5

STRETCH (○)
OFF by default
Back to the spot, fires the GARDEN POLLEN only if the camera sees our CELL still **up**.

Times are from S1 with no start delay. A start delay makes it all later. From S2 it first drives about 5 s to the shot spot.

PARKING ALWAYS WINS: at the Park-by time (24 s, or earlier if the drive is long) it stops what it is doing and parks.

During AUTO: hands off

- **Don't touch** the robot or the gamepads. The gamepads do nothing in AUTO.
- **Don't press STOP.** All motors stop at once and the robot won't park. Emergencies only.
- It waits or skips a step? **Let it.** Those are its own rules (middle column).
- Watch the `Step` line to see what it is doing.

After AUTO: TeleOp

1. The 30 s AUTO timer ends AUTO. The robot holds its park until then.
2. **BioBuzz TeleOp** is already selected for you: press **INIT**.
3. `Alliance` line: AUTO already set our colour. Check it anyway.
4. Field-centric on? The screen says `Start heading: from AUTO`. Don't press OPTIONS.
5. Don't touch the robot. ▶ **START** when TeleOp starts.

If this happens in AUTO

If...	the robot...
No CELL in view	fires the pre-loads on the planned heading.
Our CELL is already down (HIVE tipped)	never fires into it (G417). No GARDEN; it parks.
Windmill not empty after shooting (e.g. a jam)	skips the GARDEN (G407: max 4 balls) and parks.
Blocked 1 s before the park (e.g. by our partner)	drops that step and goes to park.
Blocked while parking	waits and tries again until AUTO ends. Already partly in the zone? It stays.

You: hands off. Tell the coach after the match: the log has every step.

The Step line during AUTO

Step says	It is...
`DELAY`	waiting (start delay)
`SHOT SPOT · AIM … · FIRE …`	shooting the pre-loads
`GARDEN …`	in the GARDEN
`BACK TURN · AIM (GARDEN POLLEN)`	stretch (only when ON)
`PARK: …`	driving to the slot
`PARKED B / PARKED A`	**done ✓**
`FALLBACK DONE (LEAVE only)`	Pedro was not ready

PRE-MATCH CHECKLIST

- ☐ **Partner agreed:** who has S1 · A or B · our GARDEN · our lane
- ☐ **Op mode:** BioBuzz AUTO RED / BLUE = our colour
- ☐ **4 pre-loads**, one in each pocket
- ☐ **Robot at S1 or S2:** intake against the wall, centred, square
- ☐ **INIT**, then hands off the robot
- ☐ Title line = **our colour**
- ☐ **Start** and **Park slot** as agreed
- ☐ **Shoot ON, GARDEN ON** · stretch, delay, park by as agreed
- ☐ **Route check: OK** · **Pedro: ready** · Camera not NOT FOUND
- ☐ **Place** line = where the robot is
- ☐ Last look, then ▶ **START** when the match starts. Hands off.